# Alcohol Exposure and Cell State in the Served *Pseudomonas putida* Data

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KG 4.0 serves 32 *P. putida* datasets. Fifteen of them carry an alcohol growth or tolerance arm, or a perturbation of cell state, and Table 1 gives each one's genotype, environment, phenotype and genome as the served store holds them, with the environment axis broken out in Table 3. Every count in both tables is read off the store rather than off a loader docstring (Sec. 1).

Two findings come before the table. First, no served *P. putida* dataset measures isobutanol. A search of every loader module under `torchcell/datasets/pputida/` returns no match, and the two isobutanol datasets in the release are Lopez 2024, which is *S. cerevisiae* (Sec. 2). Second, the alcohol arms that do exist split into two kinds that a query must not merge: the transposon compendia feed an alcohol as the sole carbon source, which measures catabolism, while Lim 2022 and Lim 2025 apply one as a stressor (Sec. 3).

Four of the fifteen rows are transposon screens, and they are the largest by three orders of magnitude. Sec. 5 sets out what a transposon record means here, since the genotype is deliberately a gene-level aggregate rather than a clone and the phenotype deliberately avoids the fitness class.

Sec. 6 turns to the task these rows were surveyed for: predicting the final design-build-test-learn round of the Carruthers isoprenol campaign from its earlier rounds. It first lists all seventeen served datasets in which isoprenol appears, grouped by the role it plays, because only one of those roles is a label for this task and the largest block by far is another. It then names the one non-Carruthers dataset that transfers, and records a blocker, which is that the cycle label is absent from every served Carruthers record. Sec. 7 closes with what the schema cannot yet hold.

*Section status:* ✓ ready ■ author-reviewed, keep stable ✗ not done, agents may edit freely

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# 1 Where the numbers come from ✗

Two layers, and the split is what makes the tables auditable. Every count is probed from the served store. The row set and the last column of Table 1 are curation, held as data in the generating script so the judgment is versioned with the numbers it sits beside.

**The stores read.** The probe reads the dev build tree, `$DATA_ROOT/data/torchcell/<dataset>/processed/`, which is the tree a loader writes. It is not the knowledge-graph tree under `database/data/torchcell/`, which the build user owns and which is read-only here. The two were checked against the build manifest of release 2026.10.17-4295f7ca: every record count in Table 1 equals the manifest's `n_experiments` for that dataset, which is the evidence that the probed stores are the served ones.

**Records are pickled, not JSON.** An LMDB value is a pickle, so a probe reads it with `pickle.loads`. Reading one as UTF-8 fails on the pickle protocol byte, which is how this was found.

**Constants are interned.** An `Environment`, a medium and a compound are each stored once in a sibling `processed/interned/` LMDB, and a record carries `{"$ref": <sha256>, "name": <hint>}` in their place. A probe that does not splice these back reads every environment as an opaque reference. The script splices recursively before touching any field.

**Distinct environments are counted by value.** The interned table is the tempting route to the count and it is the wrong one: it holds every interned constant, environments and media and compounds alike, and nothing guarantees each entry is still referenced. So the probe reads every record, splices its environment and hashes the spliced object, which makes two references to the same condition one environment. That costs a full scan of 2.7 million records and it is exact. The result is cached in `experiments/database/results/pputida_cell_state_probe.json` so rebuilding the document does not rescan.

**The genome is on the reference, not the experiment.** Every row's assembly in Table 1 is read from `reference.genome_reference`. The experiment-level `genome_reference` is null in all fifteen datasets. That is by design rather than a defect: the genome never gates a record, so a record does not carry one.

All fifteen pin the same assembly, `GCA_000007565.2`, while the strain on that assembly varies: KT2440 for the screens, and the engineered backgrounds IY1449b, IPL400 and CJ522 for the production and tolerance work. An engineered strain's episomal cassettes are carried on the genotype as perturbations rather than folded into the assembly, which is why one accession covers every row.

**What the probe cannot see.** Whether a transposon pool was barcoded, and whether an alcohol was fed as a carbon source or applied as a stressor, are properties of the experiment that no stored field distinguishes. Both are carried as curation, the first in the italic line under the genotype column and the second in prose in Sec. 3.

# 2 There is no served *P. putida* isobutanol data ✗

This is the first thing to establish, because the release does contain isobutanol datasets and they are the wrong organism.

**The loaders.** A case-insensitive search for *isobutanol* across the fifteen loader modules under `torchcell/datasets/pputida/` returns no file.

**The release.** The two isobutanol datasets in KG 4.0, `IsobutanolScreenLopez2024` and `IsobutanolValidatedLopez2024`, resolve in the build manifest to `torchcell/datasets/scerevisiae/lopez2024.py`. They are *S. cerevisiae*.

**The Fitness Browser deposit.** The deposit's own database carries no isobutanol experiment for any organism, so the gap is upstream of our ingest rather than something a loader dropped.

**One false positive to know about.** A text search for *isobutyr* in the Borchert 2024 environment agents hits DL-3-aminoisobutyric acid, which is an amino acid fed as a nitrogen source. Matching on compound name rather than on structure is what produces that hit; the InChIKey does not.

The closest served alcohols are the nine alcohol and diol carbon sources the transposon compendia carry, 2-methyl-1-butanol among them, and the butanol and isopentanol arms of Lim 2022. None is isobutanol. A query that needs isobutanol in *P. putida* is a query for a dataset we have not ingested, not one for a filter over what is served.

# 3 The fifteen rows ✗

Table 1 is the row table and Table 2 its totals. Two distinctions decide whether a row answers a given question, and neither is a stored field, so both are stated here rather than read off the table.

**Catabolism is not tolerance.** In the transposon compendia the alcohol arms are carbon-source perturbations: the alcohol is the sole carbon source at 10 mM, and the fitness value says whether disrupting a gene impairs growth *on* that alcohol. In Lim 2022 and Lim 2025 the alcohol is a stressor added to a glucose medium, and the readout says whether the strain tolerates it. A query that filters on the compound alone returns both and they answer different questions.

**Scale is not information.** Almost every record in Table 2 is in the two pooled compendia, which that table reports as a share, because a pooled screen multiplies one condition by several thousand genes. The engineering rows at the bottom of the table hold single digits of records and carry the phenotype the strain-design work actually targets. The record count is a cost estimate, not a ranking.

**The environment axis is the narrow one.** Fifteen datasets, and the distinct environment counts in Table 3 are dominated by the two compendia's carbon and nitrogen source panels. Outside them the condition axis is one or two environments per dataset, so almost every row is a genotype series measured in a single condition.

**Table 1:** The served *P. putida* datasets bearing on alcohol exposure or cell state, ordered by record count. Every column but the last is read off the served store (Sec. 1). *Genotype* gives the perturbation types the records carry, with what the probe cannot see added in italic. *Env.* is the number of distinct `Environment` objects, counted by value over a full scan, and Table 3 breaks each one into its media, factors and agents. *Phenotype* is the stored class, then its measurement type and assay type. *Genome* is the assembly the reference record pins; the experiment-level `genome_reference` is null throughout, by design, since the genome never gates a record.

| Dataset | Records | Genotype | Env. | Phenotype | Genome | Why it is here |
|---|---|---|---|---|---|---|
| **RbTnseq Borchert2024** | 1,466,920 | transposon insertion *gene-level aggregate, one gene per record* | 155 | **environment_response** log2_ratio / pooled_competitive_growth_barcode | KT2440 GCA_000007565.2 | The RB-TnSeq compendium, and it carries Thompson 2020's alcohol panel. Nine alcohol and diol carbon sources, 2-methyl-1-butanol among them. These arms are growth ON the alcohol as the sole carbon source, so they measure catabolism, not tolerance to a solvent stress |
| **FitnessBrowser Putida** | 1,118,052 | transposon insertion *gene-level aggregate, one locus per record* | 146 | **environment_response** log2_ratio / pooled_competitive_growth_barcode | KT2440 GCA_000007565.2 | The Fitness Browser deposit over the same alcohol panel, on 4,778 loci, and it is the one compendium that carries a moderated t statistic per record alongside the fitness value |
| **FitnessContrast Borchert2023** | 64,853 | transposon insertion | 12 | **t_statistic** rbtnseq_gene_fitness_two_sample_t_test / None | KT2440 GCA_000007565.2 | The significance test of a lignin-related stressor arm against its reference arm, stored as the t statistic rather than as a fitness value |
| **EnvMetalTnseq Royet2025** | 21,583 | transposon insertion *non-barcoded mariner pool, gene-level aggregate* | 4 | **environment_response** log2_ratio / other | KT2440 GCA_000007565.2 | Cobalt, copper, zinc and cadmium at sub-inhibitory dose over 12 generations. A metal stress rather than a solvent one, and the cleanest served example of a graded cell-state perturbation on a transposon pool |
| **GradedGene Essentiality Royet2025** | 11,458 | transposon insertion | 2 | **essentiality_call** None / None | KT2440 GCA_000007565.2 | The unstressed essentiality baseline the metal contrasts above are read against, so the two rows are used together or not at all |
| **RbTnseq Borchert2023** | 10,824 | transposon insertion | 32 | **environment_response** log2_ratio / pooled_competitive_growth_barcode | KT2440 GCA_000007565.2 | The loci Borchert 2023 scored that the compendium eliminated. Disjoint from the compendium row by construction, so it adds fitness values rather than duplicating them |
| **CampaignProteome Carruthers2025** | 465 | crispr interference, heterologous pathway | 1 | **protein_abundance** dia_nn_top3_percent_of_proteome_mean / None | IY1449b GCA_000007565.2 | Cell state under CRISPRi knockdown across the six automated design-build-test-learn cycles, as a proteome panel. The paired feature table for the isoprenol titer campaign |
| **PutidaPrecise321 Lim2022** | 321 | released identifier | 118 | **expression_tpm** rnaseq_tpm / None | CJ522 GCA_000007565.2 | The only served butanol exposure, in the compendium's Fuel project, beside an isopentanol arm, a three-point growth-phase series, a multistress panel of salt, imipenem and peroxide, an ionic-liquid adaptation series and two bioreactor runs. The broadest cell-state axis in the served set |

*Table 1, continued*

| Dataset | Records | Genotype | Env. | Phenotype | Genome | Why it is here |
|---|---|---|---|---|---|---|
| **IsoprenolSelection Menasalvas2025** | 58 | crispr interference, deletion strain, heterologous pathway | 1 | **environment__ response** categorical / biosensor__ readout | KT2440 GCA__ 000007565.2 | Isoprenol production coupled to growth through a pyrF biosensor, so the readout is a categorical enrichment call rather than a titer |
| **Proteome Menasalvas2025** | 6 | deletion strain, heterologous pathway, locus variant call, site variant call | 2 | **protein__abundance** dia__protein__counts__sum__ mean / None | KT2440 GCA__ 000007565.2 | Growth phase against production phase over 2,090 loci, with the two phases stored as distinct environments rather than as one condition |
| **ProteomeFold ChangeLim2025** | 4 | deletion strain, locus variant call, site variant call | 2 | **protein__fold__change** dia__nn__top3__log2__fold__ change__welch__two__ sample__t__test / None | IPL400 GCA__ 000007565.2 | The proteome response to isoprenol stress, as a log2 fold change with its Welch test, which is the closest served analog of a solvent tolerance signature |
| **IsoprenolTolerance Lim2025** | 3 | deletion strain, locus variant call | 2 | **environment__ response** log2__ratio / liquid__od__ growth | KT2440 GCA__ 000007565.2 | The only served alcohol tolerance growth phenotype: specific growth rate of a tolerized lineage against wild type at the same alcohol dose |
| **ProteomeLim2025** | 3 | deletion strain, locus variant call, site variant call | 1 | **protein__abundance** dia__nn__top3__log2__mean / None | IPL400 GCA__ 000007565.2 | Absolute abundances on the same tolerized lineages as the row above, so the tolerance phenotype has a proteome beside it |
| **MetaboliteGrowth Phase Menasalvas2025** | 3 | deletion strain, heterologous pathway, locus variant call, site variant call | 1 | **metabolite__level** lc__ms__intracellular__ concentration__uM / None | KT2440 GCA__ 000007565.2 | Intracellular pool sizes in growth phase, the metabolite half of the same phase contrast |
| **Metabolite ProductionPhase Menasalvas2025** | 3 | deletion strain, heterologous pathway, locus variant call, site variant call | 1 | **metabolite__level** lc__ms__intracellular__ concentration__uM / None | KT2440 GCA__ 000007565.2 | The paired production-phase pools, which is the environment the titer campaigns actually sample |

**Table 2:** Totals over the rows of Table 1, generated with them.

| Quantity | Value |
|---|---|
| Datasets | 15 |
| Records | 2,694,556 |
| Records in the two pooled transposon compendia | 2,584,972 |
| Their share of all records | 95.9% |
| Distinct environments, summed over datasets | 480 |
| Distinct phenotype classes | 7 |
| Distinct genotype perturbation types | 7 |
| Distinct genome assemblies | 1 |

# 4 Reading the environment column ✗

The `Environment` object served in all fifteen datasets carries the same seven fields: `media`, `temperature`, `perturbations`, `aerobicity`, `duration_hours`, `duration_generations` and `dilution_rate_per_hour`. Each perturbation names a factor from a controlled vocabulary and an agent resolved to a compound identity with an InChIKey and a PubChem identifier, so an alcohol arm is queryable by structure. Table 3 breaks each row into those parts.

Two caveats on that column, and both limit what a query over it can claim.

**Lim 2022's media are condition labels, not recipes.** The compendium releases a condition name per sample and not the medium, the additions or the timing, so the media count for that row counts labels. A join on medium composition across that row and any other is not available.

**A temperature can be a typed absence.** Lim 2025 reports no incubation temperature for its tolerance assay, its evolution flasks or its proteome cultures, so the field carries a `ProvenanceGap` naming where it was looked for rather than a guessed 30 °C. Table 3 prints that absence as a value, since treating it as missing data is what turns a recorded gap into an invented number.

**Table 3:** The environment axis of each row in Table 1. *Env.* is distinct `Environment` objects and *Media* distinct media names. *Factors* counts the environment perturbations carrying each factor, summed over records, so a dataset of many records shows large counts for a few conditions. *Agents* is distinct compounds resolved to an InChIKey and a PubChem identifier, which is what makes an alcohol arm queryable by structure rather than by string. *Temperature* prints every value the store holds, a typed absence included.

| Dataset | Env. | Media | Factors (perturbations carrying each) | Agents | Temperature |
|---|---|---|---|---|---|
| RbTnseqBorchert2024 | 155 | 4 | carbon 974,792, nitrogen 492,128, dropout 89,908, small mol. 492,128 | 108 | 30 °C |
| FitnessBrowserPutida | 146 | 3 | carbon 621,140, nitrogen 496,912, dropout 90,782, small mol. 277,124 | 97 | 30 °C |
| FitnessContrast Borchert2023 | 12 | 1 | carbon 64,853, small mol. 64,853 | 13 | 30 °C |
| EnvMetalTnseqRoyet2025 | 4 | 1 | small mol. 21,583 | 4 | 28 °C |
| GradedGeneEssentiality Royet2025 | 2 | 2 | none | 0 | 28 °C |
| RbTnseqBorchert2023 | 32 | 1 | carbon 10,824, small mol. 9,417 | 13 | 30 °C |
| CampaignProteome Carruthers2025 | 1 | 1 | small mol. 465 | 1 | 24 °C |
| PutidaPrecise321Lim2022 | 118 | 115 | carbon 15 | 3 | not reported |
| IsoprenolSelection Menasalvas2025 | 1 | 1 | pH 58, small mol. 116 | 2 | not reported |
| ProteomeMenasalvas2025 | 2 | 1 | pH 6, small mol. 6 | 1 | not reported |
| ProteomeFoldChange Lim2025 | 2 | 1 | small mol. 2 | 1 | not reported |
| IsoprenolTolerance Lim2025 | 2 | 1 | small mol. 3 | 1 | not reported |
| ProteomeLim2025 | 1 | 1 | small mol. 3 | 1 | not reported |
| MetaboliteGrowthPhase Menasalvas2025 | 1 | 1 | pH 3, small mol. 3 | 1 | not reported |
| MetaboliteProduction PhaseMenasalvas2025 | 1 | 1 | pH 3, small mol. 3 | 1 | not reported |

# 5 What a transposon record means here ✗

Four rows of Table 1 are transposon screens and they hold almost every record in it, so the representation is worth stating exactly.

**The experiment.** A randomly barcoded transposon library is a pool in which each cell carries one transposon inserted at a random genomic position, with a random DNA barcode in the insert. An insertion inside a gene disrupts it, so each cell is a loss-of-function mutant of one gene. The pool grows competitively in a condition, barcodes are sequenced at time zero and at the end, and a gene's fitness is the log2 ratio of its barcode abundance after growth over time zero, averaged across that gene's insertion strains and normalized so the typical gene sits at zero.

**The genotype is a gene, not a clone.** Each record carries one `TransposonInsertionPerturbation` with `insertion_position`, `insertion_strand` and `barcode` left null. Those nulls are the representation, not missing data: the stored value is the gene-level aggregate over every insertion mutant of that gene, which is the quantity the release reports. Royet 2025 is the case that makes this unavoidable, since its mariner library is not barcoded at all and a clone is unresolvable in principle.

**The phenotype is not the fitness class.** These records use `EnvironmentResponsePhenotype`, whose measurement type is `log2_ratio` and whose assay type is `pooled_competitive_growth_barcode`. `FitnessPhenotype` would be the obvious class and it is wrong here: `validate_fitness` clamps a non-positive value to 0.0, which would erase every fitness defect in the compendium, since a defect is exactly a negative log2 ratio. The signed ratio therefore goes on the environment-response class.

**Reliability travels with the value.** The Fitness Browser row carries a moderated $t$ in `environment_response_t_statistic`, and the loader derives a standard error as $\sqrt{(f/t)^2 - 0.1^2}$, where $f$ is the fitness value. That makes it the one compendium row where a per-record weight is available, which matters for any model that should down-weight a gene measured in a sparse pool.

**Both compendia do measure isoprenol, as a substrate.** The alcohol panel includes isoprenol itself as the sole carbon source at 10 mM, in MOPS minimal medium. Table 4 gives the record counts, which are the largest isoprenol block in the served data by three orders of magnitude. Each record is one gene's fitness when the pool has nothing but isoprenol to grow on, so the arm names the catabolism genes: a gene whose disruption impairs growth on isoprenol is a gene the host uses to consume it.

That makes the arm an engineering target in its own right and in the opposite direction from a titer campaign. A producing strain that also catabolizes its product loses yield to its own metabolism, and deleting a catabolism gene is how that is stopped, which is the design de Siqueira 2025's parent strain already embodies. The compendia are where the candidate genes are, genome-wide, without a new experiment.

**For titer prediction they are features, not labels.** The substrate arm does not make these rows training data for Sec. 6's task, and three differences are why. The readout is competitive growth, not product titer, and a gene that raises titer can be fitness-neutral or fitness-costly. The perturbation differs in kind: a transposon insertion is a full disruption while CRISPRi is partial repression of a gene that is still present, so the sign may agree but the magnitudes share no scale. And the chassis differs, since the libraries are wild-type KT2440 and carry neither the mevalonate pathway nor its flux burden.

What they can supply is a descriptor. A gene's fitness across the compendium's many carbon and nitrogen sources is a cheap, genome-wide measure of how essential or buffered that gene is, available for nearly every CRISPRi target without a new experiment. *Hypothesis, untested:* that descriptor improves last-round titer prediction over the proteome features alone. Nothing here measures it, and it should be checked against a proteome-only model before it is relied on.

# 6 Predicting the last Carruthers round ✗

The task these rows were surveyed for: predict the final design-build-test-learn round of the Carruthers isoprenol campaign from its earlier rounds, using whatever other isoprenol data is served. Carruthers is a collaborator, so the campaign's own data is available; the question is which single dataset from outside it helps.

## 6.1 What the served data holds on isoprenol ✗

Seventeen of the 32 served *P. putida* datasets carry isoprenol, in four distinct roles, and Table 4 lists them all. The grouping matters more than the count, because the roles answer different questions and only one of them is a label for this task:

- **As the measured product**, in five titer campaigns. These are the only candidate labels. Kang 2026 is in this group on a technicality worth knowing: it is an isoprenyl acetate campaign built on the isoprenol chassis, and 7 of its 26 records carry isoprenol as the product while 19 carry the ester.
- **As the sole carbon source**, in both transposon compendia, which is the largest isoprenol block in the served data and measures catabolism rather than production (Sec. 5).
- **As a stressor**, in the three Lim 2025 datasets.
- **As the pathway the strain carries**, in the proteome, metabolite and selection datasets, which measure something else on a strain whose genotype names the cassette. These join to a campaign by strain rather than carrying a titer.

Of the four titer campaigns outside Carruthers, three are ruled out before any comparison of conditions. Banerjee 2024 holds 20 records across five condition families, and none of its strains is a CRISPRi knockdown, so there is no target set to share. de Siqueira 2025 holds 10 records on acetate-tolerized isolates, again not CRISPRi. Kang 2026's 7 isoprenol records are controls in an ester campaign. That leaves one.

## 6.2 The dataset: Yunus 2026 ✗

`IsoprenolTiterYunus2026` is the answer, and Table 5 is why. Yunus 2026 (doi:10.1016/j.ymben.2025.11.007) selects CRISPRi knockdown targets by FluxRETAP against intuition, in the engineered isoprenol-producing strain IY1452, and Carruthers is a co-author. Four things line up and one does not.

**The target sets nearly coincide.** Every target in the Yunus campaign but one is also a Carruthers target (Table 5). That is what makes the dataset unusual: it is not a different experiment that happens to share a product, it is a second measurement of nearly the same target set.

**The readout is the same quantity.** Same product, same unit, same quantification method, same 48 hour culture. No unit conversion and no method bridge stands between the two campaigns. The replicate structure is the one readout axis that differs, and the difference is in what was released rather than in what was run. Carruthers reports a per-strain count, 2 to 6 biological replicates, and every record carries a sample standard deviation. Yunus releases one number per strain with the replicate count given only as a 3-to-6 range and the dispersion only as error bars, so the loader stores the conservative lower end of that range and records the missing dispersion as a typed gap. The uniform 3 in Table 5 is therefore a resolved lower bound, not a reported count, and a weighting scheme must not read it as one.

**The chassis and the modality are the same.** Both repress a native gene by CRISPRi in an isoprenol-producing strain, so a target's effect means the same thing in both.

**The paired features exist.** `CrispriPanelProteomeYunus2026` carries an absolute Top3 proteome over 253 accessions on the same strains, which is the same feature modality as `CampaignProteomeCarruthers2025` in Table 1. A proteome-to-titer map can be fit on both.

**The conditions differ, and this decides how it is used.** Carruthers runs at 24 °C in M9-NREL with 20 g/L glucose; Yunus runs at 30 °C in plain M9 salts. The titer ranges differ accordingly, and the Yunus records carry no uncertainty, because only one number per strain was released and the loader records that as a typed gap rather than inventing a dispersion.

So Yunus is not extra training rows for absolute titer. Two uses survive the condition difference, and both are worth more than 125 rows would be:

- **A held-out condition.** If a model trained on the earlier Carruthers cycles ranks targets correctly in Yunus, the ranking is a property of the biology rather than of the campaign. Test rank agreement, not absolute error, since the scales differ.
- **A reproducibility bound on the label.** Two independent measurements of nearly the same target set say how much of the titer variation is real. That ceilings what any model can achieve on this data, and it is the first thing to compute.

The runner-up is `IsoprenolSelectionMenasalvas2025` in Table 1, the only other CRISPRi and isoprenol pairing. Its readout is a categorical biosensor enrichment call rather than a titer, so it is a weaker label and not the first choice.

### 6.3 The blocker: the cycle label is not in the graph ✗

The task needs a cycle label, to train on the earlier rounds and test on the last one. No served Carruthers record carries one. Table 6 searches every record in all four Carruthers stores for a DBTL token anywhere in its serialization and finds none, and `screen_id` is empty throughout.

The loader's record identity is one per construct and DBTL cycle, so the cycle is implied by the record set and was never written onto a record. From the graph alone the final round cannot be separated from the earlier ones, which means the task as stated cannot be set up from a query.

It is recoverable. The raw mirror's Dryad culture metadata carries a `DBTL_Cycle` column, and Table 7 gives its distribution. The join is on the construct name, so recovering the label is a preprocessing step rather than a re-ingest. It is still a loader gap worth filing: a campaign whose entire structure is the cycle should not lose the cycle on ingest.

**Table 4:** Every served *P. putida* dataset in which isoprenol appears, and the role it plays, grouped by role. *Isoprenol* counts the records in which it appears in that role. A compound is matched on its InChIKey rather than its name, so a record spelling it 3-methyl-3-buten-1-ol is found and DL-3-aminoisobutyric acid is not; the pathway role is the exception, since a heterologous cassette names the product it makes rather than carrying its structure. The 15 served *P. putida* datasets in which isoprenol never appears are not listed.

| Dataset | Records | Isoprenol | What it measures |
|---|---|---|---|
| **Isoprenol as the measured product** | | | |
| **IsoprenolTiter Carruthers2025** | 502 | 502 | six automated DBTL cycles of CRISPRi arrays |
| **IsoprenolTiterYunus2026** | 125 | 125 | FluxRETAP targets against intuition in IY1452 |
| **IsoprenylAcetateTiter Kang2026** | 26 | 7 | an isoprenyl acetate campaign on the isoprenol chassis; both products in one store |
| **IsoprenolTiterBanerjee2024** | 20 | 20 | GSMM-guided campaign; every number read from the article prose |
| **IsoprenolTiterDe Siqueira2025** | 10 | 10 | acetate-tolerized isolates of the catabolism-deficient parent |
| **Isoprenol as the sole carbon source** | | | |
| **FitnessBrowserPutida** | 1,118,052 | 9,556 | the same panel in the Fitness Browser deposit, with a moderated t per record |
| **RbTnseqBorchert2024** | 1,466,920 | 9,464 | genome-wide fitness for growth on isoprenol, 10 mM, in the Thompson 2020 alcohol panel |
| **Isoprenol as the stressor in the medium** | | | |
| **IsoprenolToleranceLim2025** | 3 | 3 | growth rate of a tolerized lineage against wild type at the same dose |
| **ProteomeLim2025** | 3 | 3 | absolute abundances on those tolerized lineages |
| **ProteomeFoldChange Lim2025** | 4 | 2 | the stress proteome as a log2 fold change with a Welch test |
| **Isoprenol as the pathway the strain carries** | | | |
| **CampaignProteome Carruthers2025** | 465 | 465 | the proteome beside the titer campaign, cycle by cycle |
| **IsoprenolSelection Menasalvas2025** | 58 | 58 | biosensor enrichment; the titer it proxies is not measured per clone |
| **ProteomeCarruthers2025** | 21 | 21 | absolute proteome of the campaign strains |
| **ProteomeFoldChange Carruthers2025** | 16 | 16 | fold change against the per-cycle control |
| **ProteomeMenasalvas2025** | 6 | 6 | growth phase against production phase |
| **MetaboliteGrowthPhase Menasalvas2025** | 3 | 3 | intracellular pools in growth phase |
| **MetaboliteProductionPhase Menasalvas2025** | 3 | 3 | the paired production-phase pools |

**Table 5:** The two served isoprenol titer campaigns, axis by axis, every value read off the store. The product is the same compound in both, isoprenol at InChIKey CPJRRXSHAYUTGL-UHFFFAOYSA-N and PubChem 12988. *Targets shared* is the intersection of the two CRISPRi target sets, counting the CRISPRi half of the genotype only, since a Carruthers record also carries the heterologous pathway.

| Axis | Carruthers 2025 | Yunus 2026 |
|---|---|---|
| Records | 502 | 125 |
| CRISPRi targets | 135 | 121 |
| Targets shared by both campaigns | 120 | |
| Genotype perturbation types | crispr interference, deletion strain, gene addition, heterologous pathway | crispr interference |
| Medium | M9-NREL + 20 g/L glucose (10 mM ammonium sulfate + Teknova T1001; Carruthers 2025) | M9 salts (6.78 g/L Na2HPO4, 3 g/L KH2PO4, 1 g/L NH4Cl, 0.5 g/L NaCl) |
| Temperature | 24 °C | 30 °C |
| Duration | 48 h | 48 h |
| Replicates | 2 biological_replicate, 3 biological_replicate, 6 biological_replicate | 3 biological_replicate |
| Quantification | GC-FID | GC-FID |
| Titer unit | ug/mL | ug/mL |
| Titer range | 0.00 to 771.2 | 1.59 to 1,495.0 |
| Records carrying an uncertainty | 502 of 502 | 0 of 125 |

*Targets in Yunus 2026 and not in Carruthers 2025: PP_4129.*

**Table 6:** The DBTL cycle label in the served Carruthers stores. *Naming a cycle* counts records whose full serialization matches `DBTL\s*\d` anywhere, so it is as permissive a search as the store allows.

| Store | Records | Naming a cycle | With a screen id |
|---|---|---|---|
| campaign_proteome_carruthers2025 | 465 | 0 | 0 |
| isoprenol_titer_carruthers2025 | 502 | 0 | 0 |
| proteome_carruthers2025 | 21 | 0 | 0 |
| proteome_fold_change_carruthers2025 | 16 | 0 | 0 |

**Table 7:** The same label in the raw mirror, which is the recovery path. Rows are cultures and include each cycle's controls, so these are not record counts; they establish that the label exists upstream and joins on `Line_name`.

| Cycle | Culture rows |
|---|---|
| DBTL0 | 393 |
| DBTL1 | 183 |
| DBTL2 | 189 |
| DBTL3 | 174 |
| DBTL4 | 183 |
| DBTL5 | 186 |
| DBTL6 | 189 |
| **Total** | **1,497** |

# 7 What is released and not served ✗

One group of samples bearing directly on cell state is in no served dataset, and it is worth recording because a query cannot discover its absence.

**The dropped reactor arms.** Borchert 2024 releases 332 RB-TnSeq samples and the loader stores 310. The 22 it drops are the bioreactor and adaptation samples, under the rule `reactor_process_not_representable`: they differ by dissolved-oxygen set point, batch or fed-batch feed rate, sampling time, solvent overlay and adaptation day, and the loader's position is that collapsing distinct conditions into one environment is worse than not serving them. The arithmetic confirms which samples are served, since $310 \times 4{,}732$ is exactly the record count Table 1 reports for that row.

**The residue is narrower than that rule implies.** `Environment` already carries dissolved oxygen as a physical factor, plus `duration_hours`, `duration_generations` and `dilution_rate_per_hour`, so the oxygen set point, the sampling time and the adaptation day are representable on fields that exist today. Three things remain:

- **The solvent overlay has no representation.** A search of `torchcell/datamodels/schema.py` for overlay, two-phase, biphasic, dodecane and oleyl alcohol returns nothing. This affects 3 of the 22 samples, and it is the axis closest to in-situ product recovery for an alcohol, so it is the one worth a schema discussion.
- **Fed-batch feed rate is only partly covered.** `dilution_rate_per_hour` is documented as the controlled variable of a continuous chemostat, so reading a discontinuous fed-batch feed into it is a judgment call rather than a given.
- **The other 19 samples look admissible on existing fields.** That is a reading of the schema against the release metadata and not a verified admission. It needs owner sign-off and a rerun of the admission check before anyone acts on it.

**What the Fitness Browser row leaves out.** Of the deposit's 314 *P. putida* experiments the loader stores 234. It drops the 34 with an empty `pubId`, which includes the bioreactor series, and all 46 `VillalobosEscobedo23` samples, because every one is on R2A and the media library has no entry for it. The second exclusion is worth knowing when reading that row's agent count: the stress panel and methanol arms that deposit carries are not in the served set.